Dynamic Planet Practice Test CSE 2025

Union County Vocational Technical Schools (NJ)

Oceanography

Multiple Choice Section

Choose the correct answer for each of the following questions. Each question is worth 1 point.

1. What is the boundary called when plates slide next to each other in opposite directions?
    a. Convergent boundary
    b. Divergent boundary
    c. Transform boundary
    d. Oscillating boundary
2. The mid-atlantic ridge is an example of ocean-continent divergence.
    a. True
    b. False
3. Which of these convergent boundaries cause earthquakes?
    a. Ocean-continent convergence
    b. Ocean-ocean convergence
    c. Continent-continent convergence
    d. All of the above
4. What kind of reef typically will form after a volcanic island subsides below the sea level?
    a. Fringing reef
    b. Barrier reef
    c. Atoll reef
    d. Patch reef
5. Which layer structure of the ocean is associated with rapid pressure change?
    a. Thermocline
    b. Deep ocean
    c. Surface mixed layer
    d. Pycnocline
6. What type of tide occurs when the sun and moon make right angles with each other?
    a. Neap tides
    b. Spring tides
    c. Diurnal tides
    d. Mixed semidiurnal
7. Which water collection method is ideal for collecting water in discrete amounts at a programmed depth and for observing salinity and nutrient concentrations?
    a. CTD profilers
    b. Niskin bottles
    c. Stand-alone pump systems (SAPS)
    d. Surface water sampling
8. When defining wave characteristics, which of the following terms refers to the point where the wave breaks?

a. Surf
b. Swell
c. Fetch
d. Tsunami

9. What causes the gravitational pull for tides?
    a. Moon
    b. Sun
    c. Sun and moon
    d. Mars and the moon
10. Which ion is most abundant in seawater?
    a. Magnesium
    b. Chloride
    c. Potassium
    d. Sulfate
11. What type of radiation primarily heats the ocean's surface?
    a. Longwave infrared
    b. Ultraviolet
    c. Shortwave solar
    d. Gamma rays
12. Which of the following describes the typical three-layer structure of ocean water?
    a. Thermocline, halocline, pycnocline
    b. Mixed layer, thermocline, deep layer
    c. Epipelagic, mesopelagic, bathypelagic
    d. Surface zone, oxygen minimum zone, deep sea
13. Which type of plate boundary is associated with deep ocean trenches?
    a. Transform
    b. Divergent
    c. Convergent
    d. Passive
14. Nitrate concentrations are usually higher:
    a. At the surface
    b. In the deep ocean
    c. Near coral reefs
    d. Around hydrothermal vents
15. Which of these factors *do not* affect wave size?
    a. Wind Speed
    b. Fetch
    c. Salinity
    d. Duration of wind
16. Which direction does the Coriolis effect deflect surface currents in the Northern Hemisphere?
    a. Left
    b. Downward
    c. Right
    d. Westward

17. Which of the following are not one of the five major gyres?
    a. Indian Ocean Gyre (Majid Gyre)
    b. North Atlantic Gyre (Columbus Gyre)
    c. South Pacific Gyre (Heyerdahl Gyre)
    d. All of the above are part of the five major gyres
18. What causes the formation of atolls?
    a. Coral growing over subsiding volcanic islands
    b. Earthquakes that raise coral reefs
    c. Coral reefs merging with each other
    d. Tectonic uplift exposing reef systems
19. Tsunamis are caused primarily by:
    a. Seismic activity
    b. Wind
    c. Tidal current
    d. Longshore drift
20. Rip currents flow:
    a. Out to sea perpendicular to shore
    b. Along the shore
    c. Parallel to wave crests
    d. In circular gyres
21. The Atlantic Meridional Overturning Circulation does which of the following?
    a. Brings cold water east and warm water west
    b. Brings cold water west and warm water east
    c. Brings cold water north and warm water south
    d. Brings cold water south and warm water north
22. A geostrophic balance is when which forces are in equilibrium?
    a. Pressure gradient force and coriolis force
    b. Pressure gradient force and Geostrophic wind force
    c. Coriolis force and centripetal force
    d. Centripetal force and Geostrophic wind force
23. Seawater density is affected by which of the following factors?
    a. Temperature and potassium ions and coriolis forces
    b. Temperature and salinity and pressure
    c. Salinity and geostrophic force and sodium bicarbonate ions
    d. Coriolis forces and temperature and potassium ions
24. Which of the following best distinguishes an *active* continental margin from a *passive* continental margin?
    a. Active margins are found in the middle of tectonic plates, while passive margins are at plate boundaries.
    b. Active margins are characterized by broad, flat shelves, while passive margins are deep trenches.
    c. Active margins are associated with tectonic activity like earthquakes and volcanism, while passive margins are geologically stable.

    d. Passive margins typically associated with tectonic activity like earthquakes and volcanism, while active margins are geologically stable.

25. Which of the following has a horizontal pattern of distribution, increasing in concentration near the coastline?
    a. Metals
    b. Sodium bicarbonate
    c. Oxygen
    d. Nutrients
26. Tidal resonance is affected by which of the following factors?
    a. Centrifugal force
    b. Deep ocean trenches
    c. Basin geometry
    d. Coriolis force
27. Which corer tool is ideal for studying benthic organisms?
    a. Box corer
    b. Piston corer
    c. Multicorer
    d. Gravity corer
28. What do tidal bores do?
    a. Create estuaries
    b. Decrease salinity
    c. Form barrier reefs
    d. Reverses the direction of a river
29. Which ocean circulation pattern fuels the growth of phytoplankton?
    a. Atlantic Meridional Overturning Circulation
    b. Upwelling
    c. El Nino
    d. Pacific Decadal Oscillation
30. Which water tracking tool detects ancient water courses?
    a. Ground penetrating radar
    b. Piezometers
    c. Flow meters
    d. Radar sensors

Open Ended Section

Answers should be 1-2 sentences per part.

1. Natural Disasters
    a. What is the main cause of an El Niño? (2 point)
    b. What is the difference between an El Niño episode and El Niño conditions? (2 point)
    c. What effect do warm water currents have on cold water currents? How does this affect the ecosystem? (2 points)
    d. What is the difference between an El Niño and a La Niña? (2 points)
    e. Explain how El Niño and La Niña events impact marine ecosystems, specifically the impacts on upwelling and marine biodiversity? (2 points)

2. Divergent Boundaries
    a. How do divergent boundaries occur? (2 points)
    b. In ocean-ocean divergence, what happens after fissures occur? (2 points)
    c. Name a famous plate boundary that occurred from ocean-ocean divergence? (2 points)
    d. What kind of structure is formed through continent-continent divergence? (2 points)
    e. What happens when continent-continent divergence occurs below the sea level? (2 points)

3. Waves
    a. What is the highest part of a wave called? (1 point)
    b. What is the lowest part of a wave called? (1 point)
    c. Describe what the wave amplitude is? (2 points)
    d. What is the distinguishing characteristic for a capillary wave? (2 points)
    e. What are ripples caused by? (2 points)
    f. What is a standing wave? Describe it. (2 points)

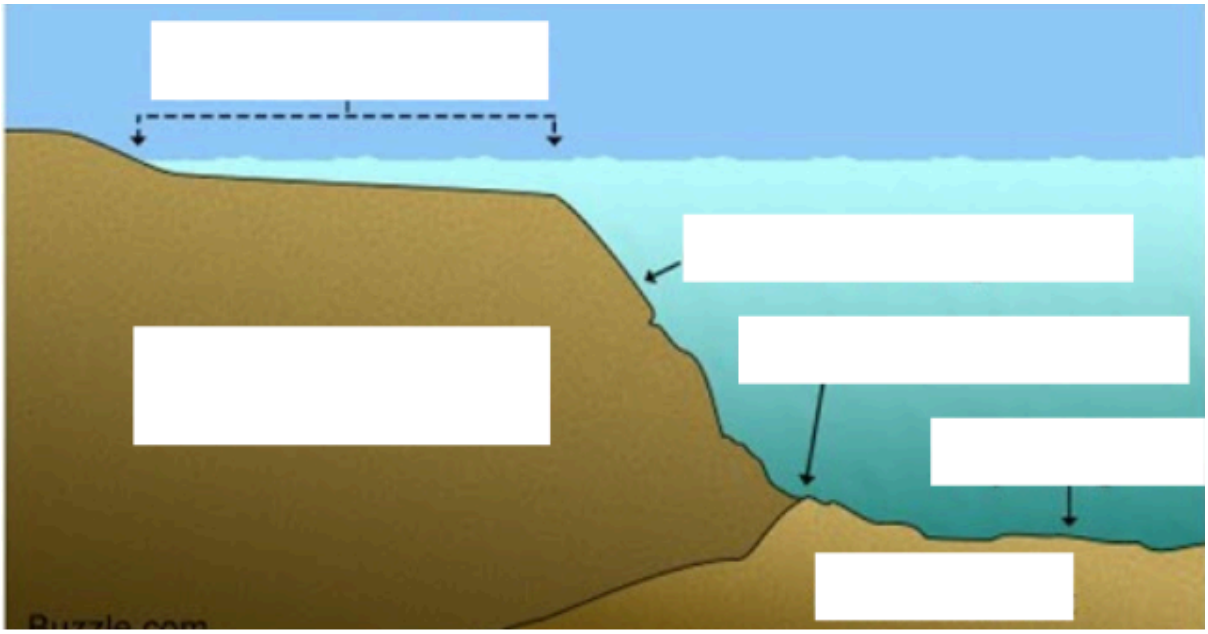
4. Ocean sediments
    a. What type of sediment is derived from a land-based origin? (1 point)
    b. Where are these sediments found? (1 point)
    c. What is the sediment from biological processes called? (1 points)
    d. Give an example of this sediment? (1 point)
    e. How do hydrogenous sediments form? (2 points)
    f. Which sediments make up most of the Earth's sediments? (2 points)
    g. What are ocean sediments which are derived from space called, such as meteorites or asteroids falling into the ocean? (2 points)

5. Surface currents
    a. Which direction are surface currents deflected to in the Southern Hemisphere? (1 point)
    b. What type of surface currents are generally found on the eastern side of ocean basins? (1 point)
    c. Describe the differences between eastern boundary and western boundary currents? (2 points)
    d. What western boundary current flows north along the east coast of the US? (2 points)
    e. What is a gyre and how are they formed? (2 points)
    f. Other than temperature, what can surface currents help regulate? (2 points)

6. Geostrophic and Ekman balances
    a. What causes the pressure gradient force? (1 points)
    b. Which direction does the pressure gradient force always act? (1 point)
    c. Why is the Ekman layer typically angled at forty five degrees? (2 points)
    d. What are the differences between an Ekman balance and a geostrophic balance? Provide three differences. (3 points)

7. Topographical features

a. Label each blank of the diagram with the correct term. (6 points)



b. Which area in the diagram supports a diverse marine habitat? (1 point)